

SQL Guidebook

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Database: Chinook

- Source: <https://www.kaggle.com/datasets/ranasabrii/chinook>
- Description: This database includes product and purchasing information for related to digital media
- Tables: Album, Artist, Customer, Employee, Genre, Invoice, InvoiceLine, MediaType, Playlist, PlaylistTrack, Track

Enter New Employee Information (INSERT INTO)

This query adds information for a new employee in the “Employee” table. This information includes the employee’s ID, first name, last name, job title, manager ID, and email.

Query

```
INSERT INTO Employee (EmployeeId, LastName, FirstName, Title, ReportsTo, Email)
VALUES (9, 'John', 'Smith', 'Sales Support Agent', 6, 'john@chinookcorp.com');
SELECT EmployeeId, LastName, FirstName, Title, ReportsTo, Email
FROM Employee
WHERE EmployeeId = 9;
```

Output

EmployeeId	LastName	FirstName	Title	ReportsTo	Email
9	John	Smith	Sales Support Agent	6	john@chinookcorp.com

Interpretation: Now, the new employee is in the Employee table and can be included in future queries.

Correct Employee Record (UPDATE)

The wrong manager ID was entered for the new employee. This query changes the manager ID to the correct ID.

Query

```
UPDATE Employee
SET ReportsTo = 2
WHERE EmployeeId = 9;
SELECT EmployeeId, ReportsTo
FROM Employee
WHERE EmployeeId = 9;
```

Output

EmployeeId	ReportsTo
-----	-----
9	2

Interpretation: Now, the correct manager ID has been set for the new employee.
Note: The manager ID is another employee ID.

Find the Top Countries Purchasing \$100+ Worth of Collections

(SELECT, SUM, FROM, GROUP BY, HAVING, ORDER BY, LIMIT)

This query identifies the total purchasing amount for each country, removes countries where the total purchasing amount is less than \$100, and returns the countries that spent the most money.

Query

```
SELECT BillingCountry, SUM(Total) AS PurchaseTotalByCountry
FROM Invoice
GROUP BY BillingCountry
HAVING SUM(Total) >= 100
ORDER BY SUM(Total) DESC
LIMIT 10;
```

Output

BillingCountry	PurchaseTotalByCountry
USA	523.06
Canada	303.96
France	195.1
Brazil	190.1
Germany	156.48
United Kingdom	112.86

Interpretation: USA is the top purchasing country.

Find Top 10 Money-Generating Collections

(WITH, AS, SELECT, SUM, FROM, JOIN, GROUP BY, ORDER BY, LIMIT)

This query finds the top 10 collections (e.g. music albums, TV series, movies, etc) that generated the most money by creating a CTE that joins all the necessary tables and calculates the total purchase amount for each collection, then arranges the purchase amounts for the top 10 collections in descending order.

Query

```
WITH MusicPurchase AS (  
  SELECT art.Name, al.Title AS Collection, SUM(i.Total) AS TotalPurchase  
  FROM Track t  
  JOIN Album al ON t.AlbumId = al.AlbumId  
  JOIN Artist art ON al.ArtistId = art.ArtistId  
  JOIN InvoiceLine il ON t.TrackId = il.TrackId  
  JOIN Invoice i ON il.InvoiceId = i.InvoiceId  
  GROUP BY al.AlbumId  
)  
SELECT *  
FROM MusicPurchase  
ORDER BY TotalPurchase DESC  
LIMIT 10;
```

Output

Name	Collection	TotalPurchase
-----	-----	-----
Lenny Kravitz	Greatest Hits	372.51
Lost	Lost, Season 2	290.18
Heroes	Heroes, Season 1	238.61
Lost	Lost, Season 1	223.65
Lost	Lost, Season 3	211.8
Battlestar Galactica	Battlestar Galactica, Season 3	202.8
Chico Buarque	Minha Historia	185.13
The Office	The Office, Season 3	170.93
Zeca Pagodinho	Ao Vivo [IMPORT]	161.74
Battlestar Galactica (Classic)	Battlestar Galactica (Classic), Season 1	157.1

Interpretation: Lenny Kravitz's Greatest Hits is the top money generating collection.

Determine High and Low Spenders

(WITH, SELECT, SUM, AS, FROM, GROUP BY, CASE WHEN, LIMIT)

This query creates a CTE that calculates the total purchase amount for each customer, then the CTE is used to determine which customers are high spenders based on whether they purchased at least \$40 worth of collections.

Query

```
WITH Purchases AS (  
    SELECT CustomerId, SUM(Total) AS TotalPurchase  
    FROM Invoice GROUP BY CustomerId  
)  
SELECT CustomerId, TotalPurchase,  
CASE WHEN TotalPurchase >= 40 THEN 'Yes'  
ELSE 'No'  
END AS HighBuyer  
FROM Purchases  
LIMIT 10;
```

Output

CustomerId	TotalPurchase	HighBuyer
1	39.62	No
2	37.62	No
3	39.62	No
4	39.62	No
5	40.62	Yes
6	49.62	Yes
7	42.62	Yes
8	37.62	No
9	37.62	No
10	37.62	No

Interpretation: Customers 5,6, and 7 are high spenders, while the rest are not.

Show the Full Name of Customers

(SELECT, CONCAT, FROM, LIMIT)

This query combines the first and last names of the customers into a single variable.

Query

```
SELECT CustomerId, CONCAT(LastName, ' ', FirstName) AS FullName
FROM Customer
LIMIT 5;
```

Output

CustomerId	FullName
1	Gonçalves Luís
2	Köhler Leonie
3	Tremblay François
4	Hansen Bjørn
5	Wichterlová František

Interpretation: Now, the full name of the customers is in one variable.

Find Customer Purchase Change Over Time

(SELECT, DATE, LAG, OVER, PARTITION, COALESCE, FROM, ORDER BY, LIMIT)

This query finds the difference in purchasing amounts between each purchase for every customer by removing the time part of the date variable, keeping track of the purchase amount of the previous purchase for each customer, and subtracting the previous amount from the next purchase amount.

Query

```
SELECT CustomerId, DATE(InvoiceDate) AS PurchaseDate, Total AS PurchaseAmount,  
LAG(Total) OVER (PARTITION BY CustomerId ORDER BY DATE(InvoiceDate)) AS PreviousInvoice,  
Total -COALESCE(LAG(Total) OVER (PARTITION BY CustomerId ORDER BY DATE(InvoiceDate)),0) AS PurchaseDifference  
FROM Invoice  
ORDER BY CustomerId, PurchaseDate  
LIMIT 10;
```

Output

CustomerId	PurchaseDate	PurchaseAmount	PreviousInvoice	PurchaseDifference
-----	-----	-----	-----	-----
1	2010-03-11	3.98		3.98
1	2010-06-13	3.96	3.98	-0.02
1	2010-09-15	5.94	3.96	1.98
1	2011-05-06	0.99	5.94	-4.95
1	2012-10-27	1.98	0.99	0.99
1	2012-12-07	13.86	1.98	11.88
1	2013-08-07	8.91	13.86	-4.95
2	2009-01-01	1.98		1.98
2	2009-02-11	13.86	1.98	11.88
2	2009-10-12	8.91	13.86	-4.95

Interpretation: Customer 1 spent less money on their second, fourth, and seventh purchases compared to the purchases that preceded them. Customer 2 spent less money on their third purchase compared to their second purchase.

Fill Missing Billing States with N/A

(SELECT, COALESCE, FROM, LIMIT)

This query identifies the rows in the “Invoice” table with billing states and adds N/A to them.

Query

```
SELECT InvoiceId, CustomerId, BillingAddress, BillingCity, COALESCE(BillingState, 'N/A'), BillingCountry, BillingPostalCode
FROM Invoice
LIMIT 5;
```

Output

InvoiceId	CustomerId	BillingAddress	BillingCity	COALESCE(BillingState, 'N/A')	BillingCountry	BillingPostalCode
1	2	Theodor-Heuss-Straße 34	Stuttgart	N/A	Germany	70174
2	4	Ullevålsveien 14	Oslo	N/A	Norway	0171
3	8	Grétrystraat 63	Brussels	N/A	Belgium	1000
4	14	8210 111 ST NW	Edmonton	AB	Canada	T6G 2C7
5	23	69 Salem Street	Boston	MA	USA	2113

Interpretation: now, every missing state is filled with N/A